Towards Compute-Aware Attention: A Skip Mechanism for Redundant Transformer Heads

Chun-Min Yu

Abstract

Transformers frequently exhibit *attention sink* behavior, where heads concentrate probability on low-semantic tokens (e.g., punctuation, separators), attempting to perform near no-op routing while still incurring full computation. We present an **attention skip mechanism** that detects such sinky heads and selectively bypasses them.

Our approach measures head-wise concentration via a max-row threshold over attention matrices and applies a head mask at runtime to skip computations without altering model outputs elsewhere.

Experiments on variety natural language processing tasks show that skipping sinky heads improves performance on GPT-2 while reducing the effective contribution of redundant heads. Analysis of the value vectors suggests that tokens receiving the most attention per row tend to have smaller value vector magnitudes compared to other tokens. Additionally, skipped heads predominantly emerge in deeper layers, where they are likely to contribute less to high-level, meaningful understanding of the input sequence.

Our findings clarify sink dynamics and demonstrate a practical path to compute-aware attention, opening opportunities for training-time attention computation early stopping, thereby enhancing model performance.

1 Introduction

The Transformer architecture has become the backbone of modern natural language processing (NLP) and computer vision models. However, recent research has revealed inefficiencies in the attention mechanism, particularly in the phenomenon known as attention sinks, where attention value disproportionately gathers on low-semantic tokens, such as punctuation or special tokens, or certain positions. This often results in redundant computations without contributing meaningful contextual information, or even degrades the performance, which will later be discussed.

Earlier studies have meticulously identified and analyzed attention sinks, showing that certain columns (often the first position or the position of low-semantic tokens) accumulate disproportionate attention, and that such positions frequently carry low-norm value vectors—explaining why high attention need not imply strong contribution. These works, however, are largely descriptive: they improve our understanding but leave the attention stack unchanged at inference.

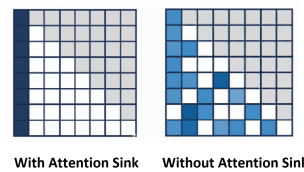

Figure 1: A brief illustration of what *Attention Sink* means

To address this inefficiency and to improve the attention architecture, we propose **Attention-Skip** mechanism, a method to selectively bypass attention heads that exhibit strong sink behavior. Our assumption underlying the attention-skip architecture is straightforward: when an attention head exhibits strong sink behavior, it tries to perform a *'no-op'* operation. Therefore, we propose to skip such heads. The mechanism is dynamic (per-step, per-head), model-agnostic, and compatible with every architecture involving attention calculation. In doing so, we turn an observational artifact into a practical efficiency primitive, reducing redundant computation while not only maintaining, but improving task performance.

In summary, our contributions are fourfold: (1) We propose a dynamic, and model-agnostic **Attention-Skip** mechanism that detects and bypasses redundant attention heads exhibiting sink behavior; (2) We empirically demonstrate that this mechanism enhances performance across diverse benchmarks, including GLUE, IMDb, and SQuAD; (3) We analyze the distribution of skipped heads across layers, revealing that redundancy concentrates in deeper layers; and (4) We provide a new interpretation of attention sinks through the lens of value-vector magnitude, offering a more complete understanding of head redundancy in Transformer architectures.

2 Related Work

Interpreting Large Attention Weights: A common reading of Transformer attention is: if entry (i, j) in the attention matrix is large, then token i "finds" token j important, and the $i \leftarrow j$ relation carries salient information. This weight-centric view underpins many interpretability and pruning methods, which rank token importance using attention scores alone. However, the output of attention is a weighted sum of value vectors, so attention weight is only half the story; the magnitude of the corresponding value vectors also determines contribution. Early norm-based analyses formalized this point and showed that attention weights by themselves can misestimate the influence (Kobayashi et al. 2020).

Value Vectors as a Proxy for Semantic Richness: A complementary assumption in practice is that tokens with larger value-vector norms tend to carry richer or more reusable features. All else being equal, they can exert stronger effects on downstream representations when attended to. Norm-aware analyses therefore recommend combining attention scores with vector norms to approximate a token's effective impact:

$$\text{Effective Impact} \;\approx\; \text{Weight} \times \text{Value}.$$

Empirically, norm cues often sharpen alignments and yield better importance estimates than weights alone.

Low–semantic tokens and value norms. Let $V \in \mathbb{R}^{B\times H\times S\times d_h}$ be per–head value tensors and $r_{b,h}(j) := \|V_{b,h,j,:}\|_2$ the ℓ_2 norm of the value vector for source position j. Denote by $\mathcal{L}$ the set of *low–semantic* tokens (e.g., punctuation, `[PAD]`, `<SEP>`, BOS/EOS, stopwords) and by $\mathcal{C}$ the content tokens. Empirically one observes a clear separation:

$$\mathbb{E}\big[r_{b,h}(j) \mid j \in \mathcal{L}\big] \;\ll\; \mathbb{E}\big[r_{b,h}(j) \mid j \in \mathcal{C}\big],$$

often by a sizable factor across layers and datasets. Intuitively, such tokens serve control or placeholder roles; their gradients are (i) averaged across many heterogeneous contexts, (ii) encouraged to be "harmless" aggregation targets for attention mass, and (iii) attenuated by normalization and residual mixing, all of which drive their learned value projections toward small magnitude.

This low-norm property explains why large attention on sinks does not imply large contribution. For a peaky row s with sink index $j^\star$ and a random baseline $R \subseteq \{1,\dots,s\} \setminus \{j^\star\}$,

$$\Delta_v \;:=\; \underbrace{\mathbb{E}\big[r_{b,h}(j^\star)\big]}_{\text{sink value}} \;-\; \underbrace{\mathbb{E}\Big[\tfrac{1}{|R|}\sum_{j\in R} r_{b,h}(j)\Big]}_{\text{non- sink baseline}} \;<\; 0,$$

so the product $\alpha_{ij^\star} v_{j^\star}$ remains small even when $\alpha_{ij^\star}$ is large. Practically, this makes low-semantic tokens effective "probability absorbers" that stabilize representations without adding much semantic content. We also note a layer trend: the gap Δ_v typically widens in upper layers where routing/stabilization heads are more prevalent, whereas early layers devote capacity to local, content-rich features.

The Existence of Attention Sink: Recent studies highlight a robust "attention sink" phenomenon: specific positions—most prominently the first token—attract disproportionately high attention from many queries, regardless of semantic content (Barbero et al. 2025; Gu et al. 2025; Xiao et al. 2023). This behavior appears across model sizes and datasets and has architectural and optimization roots (e.g., providing a universal "fallback" target to avoid over-mixing). Crucially, sinks can occur even when the first token is a normal word rather than a special delimiter. Recent work also shows that low-semantic tokens—such as punctuation marks, special tokens (e.g., `<SEP>`, `[PAD]`, $), or stop words (e.g., *the*, *a*)—consistently attract a disproportionately high amount of attention.

Let the attention output at position i be

$$o_i = \sum_j \alpha_{ij}\, v_j, \tag{1}$$

where α_{ij} are attention weights and v_j are value vectors. Sink tokens are positions that attract disproportionately large α_{ij}. Crucially, these tokens often carry low-norm value vectors (small $\|\mathbf{v}_j\|$). As a result, even when α_{ij} is large, the product $\alpha_{ij}\,\mathbf{v}_j$ contributes little to $\mathbf{o}_i$.

Intuitively, sink tokens act as probability mass absorbers: they collect attention weight that the model cannot confidently assign elsewhere, while their low value-vector magnitude prevents them from overwhelming the representation. Thus, high attention to a low-norm value vector does not imply high influence on the layer output.

3 Method

Motivation Multi-head self-attention frequently exhibits sink behavior, in which the per-row attention distribution collapses onto a single column for a large fraction of query positions. Tokens occupying these sink columns typically have low-norm value vectors $\|v_j\|$, so even when α_{ij} is large, the effective term $\alpha_{ij}\,\mathbf{v}_j$ contributes little to o_i. Moreover, because the softmax mass concentrates on the sink, the residual probability available to other tokens is small—thus their aggregate $\sum_{j\neq j^\star} \alpha_{ij} v_j$ is also trivial, even if some of those value vectors are informative.

Heads that display this pattern therefore behave almost like no-ops: they route probability mass to a fixed position while adding little information beyond the residual pathway. We leverage this observation—high attention to low-magnitude value vectors—to skip such heads at run time to help model achieve no-op. Without such skipping, if the model intends to approximate a no-op effect, it would need to assign extremely large logits to low-v tokens in order to concentrate softmax attention on them, which risks numerical instability and potential value overflow. Our mechanism avoids this inefficiency by directly bypassing these heads.

Setting Consider a Transformer block with H heads operating on input $x \in \mathbb{R}^{B\times T\times C}$. Each head forms

$$Q, K, V \in \mathbb{R}^{B\times H\times T\times d_h}, \quad d_h = C/H,$$

and computes causal attention

$$A = \mathrm{softmax}\Big(\frac{QK^\top}{\sqrt{d_h}} + M\Big) \in \mathbb{R}^{B\times H\times T\times T},$$

followed by $Y = AV$ and the standard output projection and residual addition.

Sink Indicator We diagnose sinkness inside each head after masking and softmax by measuring row-wise peakedness. For each query position i, we compute the row maximum $\max_j A_{ij}$. A row is deemed peaked if this maximum exceeds a threshold τ_{peak} (default 0.95). A head is classified as sinky for a given sequence when the fraction of non-PAD rows that are peaked exceeds a ratio ρ_{head} (default 0.8). At each step, for each batch item and head, we evaluate the peaked-row ratio. If it exceeds ρ_{head}, the head is skipped.

The procedure is fully stateless: decisions are recomputed at every forward pass from the current attention map, with no caching across steps or sequences. It is compatible with any attention structure—including KV caching—since the peakedness test operates on the current attention matrix regardless of how keys and values are formed.

For interpretability and diagnostics, attention heatmaps may be periodically saved at fixed intervals, and per-layer, per-head suppression statistics accumulated for offline analysis; these side effects do not alter computation. Hyperparameters τ_{peak} and ρ_{head} control aggressiveness: higher values reduce false positives at the cost of fewer suppressed heads, whereas lower values increase suppression but may risk attenuating legitimately sharp, content-bearing heads. In practice, conservative thresholds target genuine sinks while leaving informative heads intact.

4 Experiments

4.1 Datasets

IMDb (Large Movie Review) The IMDb dataset (Maas et al., 2011) is a document-level sentiment corpus built from movie reviews collected on imdb.com. It provides 50k labeled reviews with a balanced positive/negative split and fixed partitions: 25k for training and 25k for testing. An additional set of 50k unlabeled reviews is included for semi-/unsupervised studies, though we use only the labeled portion in our experiments. Reviews are long and stylistically varied (multi-sentence, paragraph length), making the task non-trivial due to discourse phenomena and long-range dependencies. The canonical evaluation metric is accuracy on the held-out test split; hyperparameters are selected on a small validation split carved from the training data.

GLUE Benchmark The General Language Understanding Evaluation (GLUE; Wang et al., 2018) is a multi-task benchmark designed to assess transferable language understanding. It aggregates diverse English tasks spanning single-sentence classification, sentence-pair classification, and semantic similarity:

- **CoLA**: linguistic acceptability; reported with Matthews correlation (MCC).
- **SST-2**: sentence-level sentiment; accuracy.
- **MRPC** and **QQP**: paraphrase/duplicate detection; F1/accuracy.
- **STS-B**: semantic textual similarity; Pearson/Spearman.
- **MNLI**: entailment with in-domain *matched* and out-of-domain *mismatched* test sets; accuracy.
- **QNLI** and **RTE**: question–answer/recognizing textual entailment; accuracy.
- *WNLI is often excluded due to label and protocol peculiarities.*

GLUE provides standard train/dev/test splits; test labels are withheld, and official scores are obtained via server submission. The benchmark reports per-task metrics and an aggregated GLUE score (normalized average across tasks). Its heterogeneity—varied input lengths, label balances, and reasoning types—makes GLUE a rigorous setting to evaluate whether mechanisms like attention skipping preserve generalization beyond a single task and dataset.

Our evaluation focuses on **SST-2, MNLI-m, MNLI-mm, QNLI, and QQP**. We exclude the other GLUE tasks due to limited labeled supervision, which would cause performance to depend primarily on the pretrained weights rather than task-specific adaptation. In this low-signal regime, modifying the attention architecture introduces a mismatch with the checkpoint's native inductive biases and is therefore likely to reduce accuracy.

SQuAD v1.1 and v2.0 The Stanford Question Answering Dataset (SQuAD; Rajpurkar et al., 2016, 2018) is an extractive QA benchmark built from English Wikipedia paragraphs. Each example contains a context paragraph and a question; the answer, when present, is a contiguous span in the context. SQuAD v1.1 contains only answerable questions (≈87k train, 10k dev) and is evaluated with exact match (EM) and token-level F1 on held-out splits. SQuAD v2.0 augments v1.1 with ≈53k *unanswerable* questions crafted to resemble answerable ones, yielding roughly 130k training examples in total. Systems must either predict a span or abstain ("no answer"); EM and F1 are computed using a threshold on the no-answer score, with official scoring provided by the evaluation script/server on the dev/test sets. Compared with v1.1, v2.0 emphasizes calibrated uncertainty and robustness to distractor evidence, making it a stronger test of span selection and abstention.

It is worth noting that the F1 scores reported in Table 1 are from the best validation checkpoint rather than the official test split, because evaluation on the hidden test set requires submitting code to the organizers, which is time-consuming and complex.

4.2 Experimental Design

Setup: Attention-Skip (stateless peakedness gating). At each forward pass, each head is tested for sinkness on the *current* attention map; if sinky, the head is skipped for that mini-batch only. No decisions are cached across batches/epochs.

Sink detection rule Let $A \in \mathbb{R}^{T\times T}$ denote a head's post-mask attention matrix. For each query row i, define a peaked-row indicator

$$\pi_i \;=\; \mathbf{1}\Big[\max_j A_{ij} \geq \tau_{\text{peak}}\Big],$$

and let $\mathcal{I}$ be the set of non-PAD rows. A head is labeled *sinky* for the sequence if

$$\frac{1}{|\mathcal{I}|}\sum_{i\in\mathcal{I}}\pi_i \;\ge\; \rho_{\text{head}}.$$

Unless otherwise stated, we use conservative defaults $\tau_{\text{peak}} = 0.95$ and $\rho_{\text{head}} = 0.8$ to target genuine sinks while avoiding content-bearing sharpness. We sweep both thresholds in ablations.

Runtime skipping If a head is sinky, we bypass its attention computation (softmax and value mixing) and emit zeros for its contribution $Y^{(h)}$, preserving tensor shapes so residual pathways and other heads remain unchanged. Gradients for skipped heads on those sequences are suppressed.

Value-vector mean (v-mean). We ask whether *peaky* rows (rows whose attention mass is concentrated on one column) tend to read from value vectors with unusually small magnitude. Let $V \in \mathbb{R}^{B\times H\times S_{\text{src}}\times d_h}$ be the value tensors and $w \in \mathbb{R}^{B\times H\times S_{\text{dst}}\times S_{\text{src}}}$ the attention probabilities after masking and softmax. For batch b, head h, and destination row s whose peakiness exceeds τ_{peak}, consider the valid *source positions* (attention columns, i.e., keys). When a row is peaky (most mass concentrated on one column), we name the column that gets the most probability the sink:

$$J = \{\, j \le s \;:\; \text{token } j \text{ is not PAD} \,\}.$$

The *sink index* is

$$j^\star := \arg\max_{j\in J} w_{b,h,s,j}.$$

For a source index j, let the value vector and its ℓ_2 magnitude be

$$\boldsymbol{v}_{b,h}(j) := V_{b,h,j,:} \in \mathbb{R}^{d_h},$$

$$r_{b,h}(j) := \|\boldsymbol{v}_{b,h}(j)\|_2 = \sqrt{\sum_{k=1}^{d_h} V_{b,h,j,k}^2}.$$

For each peaky row s, record the magnitude of the value vector at the sink index

$$r_{\text{sink}} \;=\; r_{b,h}(j^\star),$$

and a random-source baseline with up to 10 indices sampled uniformly without replacement

$$r_{\text{rand}} = \frac{1}{|R|}\sum_{j\in R} r_{b,h}(j),$$

$$R \subseteq J \setminus \{j^\star\}.$$

Aggregate sums and counts across steps, layers, heads, batches to form global means $\overline{r}_{\text{sink}}$ and $\overline{r}_{\text{rand}}$, and report the gap

$$\Delta \;=\; \overline{r}_{\text{sink}} - \overline{r}_{\text{rand}}.$$

If $\Delta < 0$, the sink tends to route probability to lower-magnitude value vectors, consistent with near no-op contributions.

Example. With $d_h = 64$, if $r_{\text{sink}} = 0.23$ and the random baseline averages 0.31, then $\Delta = -0.08$, indicating a lower-magnitude value at the sink.

Configurations. We compare:

- **Plain (no skip).** Standard training/evaluation.
- **Attention-Skip.** Described as above.

4.3 Baseline Models

Plain *GPT-2* (345M). A standard GPT-2 "small" model (decoder-only, learned absolute positions; HuggingFace `gpt2` weights) *fine-tuned* for each task with a randomly initialized classification head on top of the final-token representation. Optimizer, tokenization, sequence lengths, and stopping criteria follow §4.2. This baseline represents a stronger capacity point and tests whether conclusions drawn from *minGPT* transfer to a widely used pretrained backbone.

Deriving our variants. For the modified architecture **Attention-Skip**, the underlying architecture, tokenization, optimization, and all hyperparameters are kept *identical* to the corresponding Plain baseline; only the gating rule is toggled, ensuring a fair comparison.

4.4 Metrics

Task quality. We follow GLUE conventions for the selected tasks (SST-2, MNLI-m/mm, QNLI, QQP):

- **Accuracy** (SST-2, MNLI-m/mm, QNLI):

$$\text{Acc} = \frac{1}{N}\sum_{n=1}^{N} \mathbf{1}[\hat{y}_n = y_n]\,.$$

- **F1 / Accuracy** (QQP). With $\mathrm{TP}, \mathrm{FP}, \mathrm{FN}$ on the positive class:

$$\mathrm{P} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}}, \quad \mathrm{R} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}}, \quad \mathrm{F1} = \frac{2\,\mathrm{P}\,\mathrm{R}}{\mathrm{P}+\mathrm{R}}.$$

For within-paper aggregation, we report an *unofficial* macro average over the included tasks.

Gating/behavioral diagnostics.

- **Head Skip Ratio (HSR)**: percentage of total head instances skipped,

$$\text{HSR} = \frac{\#\{\text{skipped head instances}\}}{\#\{\text{all head instances}\}} \times 100\%.$$

  Reported overall and per layer.
- **Peaked-row fraction**: detector statistic $\frac{1}{|\mathcal{I}|}\sum_{i\in\mathcal{I}} \mathbf{1}[\max_j A_{ij} \ge \tau_{\text{peak}}]$, summarized across all rows in a single head to quantify sink prevalence. We set it as 0.8 as default.

5 Results

5.1 Model Performance

Table 1 reports the performance of GPT-2 and its modified variant on parts of the GLUE benchmark (SST-2, MNLI, QNLI, QQP) and on the IMDb dataset. The modified GPT-2 (GPT2 mod) consistently yields improvements on several tasks. In particular, it outperforms the baseline on SST-2 (92.9 → 93.2), QNLI (89.7 → 90.0) and QQP (69.5/88.0 → 69.5/88.5), while achieving small gains on MNLI-m

Model	SST-2	MNLI-m	MNLI-mm	QNLI	QQP (F1/Acc)	IMDb	SQuAD v1.1	SQuAD v2.0
GPT2	92.9	82.3	81.9	89.7	69.5/88.0	92.04	83.95	72.51
GPT2 mod	**93.2**	**82.5**	**82.7**	**90.0**	**69.5/88.5**	**93.20**	**84.15**	**72.78**

Table 1: Comparison of different model settings on parts of the GLUE, SQuAD v1.1, SQuAD v2.0 benchmark and IMDb dataset.

	SST-2	MNLI	QNLI	QQP
Layer 0–3	0.10%	0.00%	0.00%	0.00%
Layer 4–7	3.54%	2.90%	2.14%	5.50%
Layer 8–11	4.28%	0.75%	3.20%	3.18%
Layer 12–15	3.82%	3.40%	4.03%	2.09%
Layer 16–19	6.52%	4.57%	8.03%	2.81%
Layer 20–23	6.73%	13.9%	7.59%	11.42%

Table 2: Average head-skipping percentage out of total skipped heads by layer group for GPT-2 mod.

Token Type	Value vector L2 norm
Sink tokens	**0.89**
Random tokens	7.35
Δ_v	-6.46

Table 3: L2 Norm of Value Vectors for Different Token Types in GPT2

(82.3 → 82.5) and MNLI-mm (81.9 → 82.7). On the IMDb dataset, GPT2 mod also outperforms plain GPT2 by 1.16%. These results demonstrate that the proposed head-skipping mechanism preserves competitive performance across diverse tasks, with clear benefits in several benchmarks, suggesting that certain head computations are redundant.

Moreover, on SQuAD v1.1, GPT2 mod outperforms GPT-2 by a small margin (84.15 vs. 83.95), while also showing a noticeable improvement on SQuAD v2.0 (72.78 vs. 72.51). These results demonstrate that the proposed head-skipping mechanism not only preserves competitive performance across a wide variety of NLP classification benchmarks, but also shows consistent gains on generation tasks like SQuAD. The improvements on SQuAD v1.1 and v2.0 highlight that the model's ability to skip redundant heads aids in maintaining, and in some cases improving, performance on more complex question-answering tasks, while still offering computational savings. This suggests that certain head computations are redundant, especially in settings where no answer is expected, as demonstrated by the SQuAD v2.0 results.

5.2 Head Skipping Distribution

Table 2 presents the average head-skipping percentage across grouped layers in GPT-2 mod. Skipping activity is negligible in the lowest layers, but increases steadily with network depth. For instance, layers 8–11 skip about 4.3% of heads, while layers 16–19 and 20–23 reach 6.5% and 6.7%, respectively. The effect also varies by dataset: MNLI and QQP show particularly high skipping in deeper layers (13.9% and 11.4%, respectively). This pattern suggests that redundant or sink-like attention heads are concentrated in upper layers.

5.3 Value-Norm Comparison

Table 3 shows that value vectors associated with sink tokens (e.g., punctuation, [SEP]) have dramatically smaller magnitudes than those of typical tokens in GPT-2: the average L2 norm is 0.89 for sinks versus 7.35 for randomly sampled tokens, a gap of 6.46 (8.3× smaller). This large separation indicates that, even when attention mass concentrates on sinks, their contribution to the residual stream is inherently attenuated by low-norm value projections. The result empirically supports our compute-aware design: heads that predominantly route to sink positions carry little semantic payload and can be safely skipped with minimal risk to model expressiveness.

5.4 Summary

Overall, *GPT2 mod* demonstrates consistently stronger performance across both the GLUE and IMDb benchmarks, while simultaneously achieving measurable computational savings through dynamic head skipping. The improvements observed on SST-2, MNLI, QNLI, and QQP indicate that the proposed mechanism not only preserves but often enhances task accuracy, effectively identifying and bypassing redundant attention heads that contribute little to overall reasoning. Additionally, on **SQuAD v1.1** and **SQuAD v2.0**, *GPT2 mod* shows modest gains over the baseline, highlighting that the mechanism helps maintain competitive performance even in the more complex task such as question answering generation task. These results suggest that a substantial portion of attention computations, particularly in deeper layers, can be safely omitted without degrading model expressiveness and performance. The improvements in **SQuAD v1.1** and **SQuAD v2.0** further support the view that many upper-layer heads specialize in low-salience or repetitive routing patterns, and their bypassing contributes to more efficient computation without sacrificing accuracy.

6 Conclusion

We introduced an *attention skip mechanism* that detects attention sinks and selectively bypasses heads whose behavior is effectively near *no-op*. The method is model-agnostic, and compatible with standard Transformer implementations. Analysis of per-layer skip statistics revealed that head skipping concentrates in middle-to-upper layers, indicating that redundancy is not uniform across depth. These findings suggest that a nontrivial fraction of head computations is redundant and can be suppressed without materially degrading quality.

Implications. By converting a descriptive phenomenon (attention sinks) into a compute-aware control, our approach offers a possible path to lower-cost inference and training if hardware is integrated. It provides a plug-in mechanism for latency and energy reduction, and it supports complementary strategies such as training-time early stopping of sinky heads, KV-cache pruning, and deployment-time head gating under resource budgets.

Limitations and Future Work. Our study is limited by computational budget: we tested a single threshold configuration for the detector (peakedness τ_{peak} and head ratio ρ_{head}) and performed only light ablations. We did not explore larger models or other suitable architectures. Future work will perform comprehensive sweeps over $(\tau_{\text{peak}}, \rho_{\text{head}})$ to identify optimal configurations for this architecture, and will further extend evaluations to larger-scale models and a broader range of benchmarks.

We will also explore alternative mechanisms to achieve the same goal—enabling the model to perform a no-op. One approach is to introduce an explicit null token with $v = 0$, or to clamp designated low-v tokens to $v = 0$, combined with softmax clipping to prevent pathological concentration. Another approach is to enforce *hard* (top-1) attention in peaked rows, replacing softmax with a hardmax. These interventions may yield greater numerical stability and potentially stronger results than our current robust baseline. Absent such controls, a model can only approximate a no-op by driving the logits of low-v tokens $z_\ell \to \infty$ so that $\mathrm{softmax}(z)_\ell \to 1$, which is inefficient and prone to numerical overflow. By employing the proposed attention-skip mechanism—or analogous gating strategies—we avoid this escalation and directly suppress non-contributory heads.

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